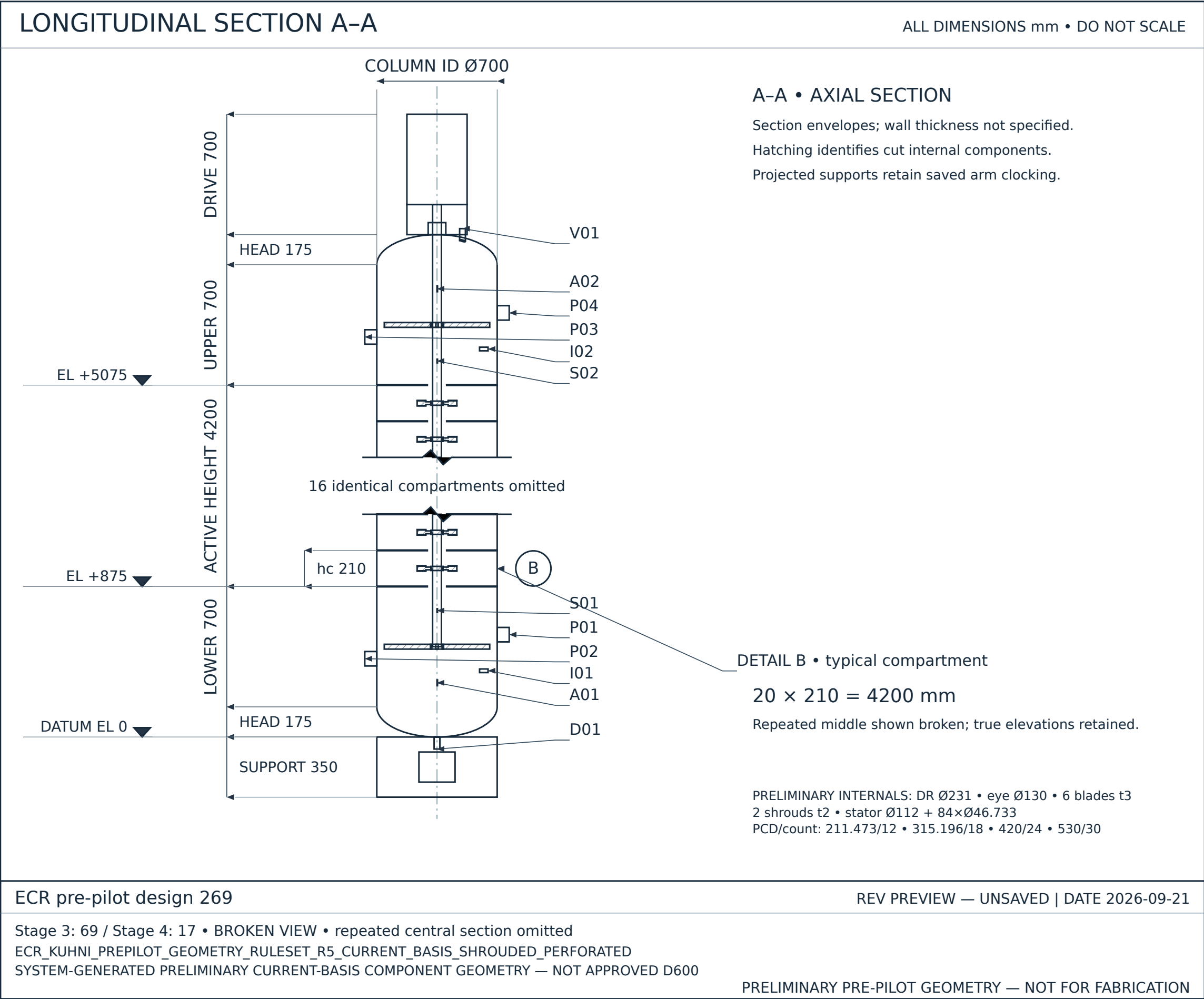
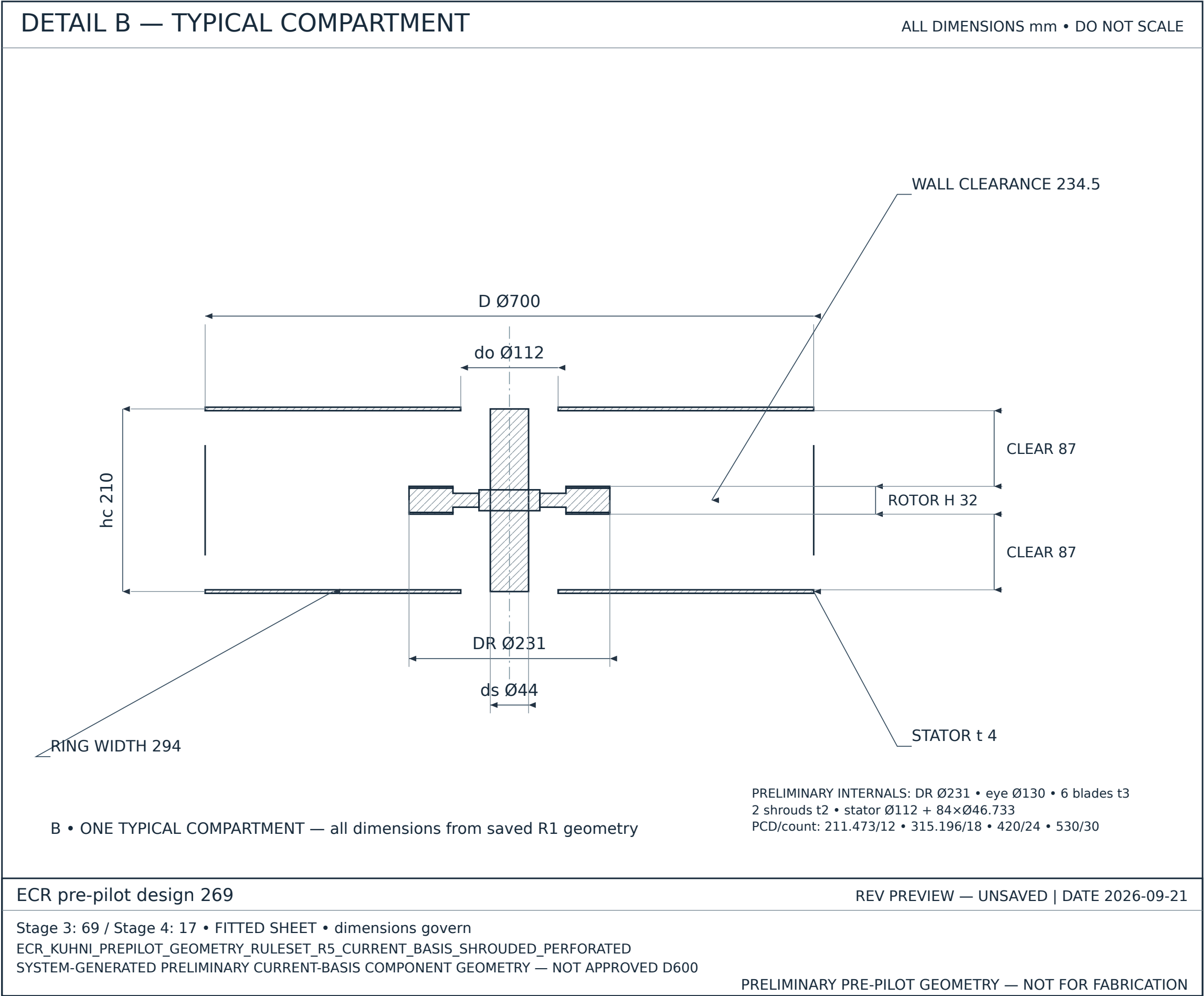
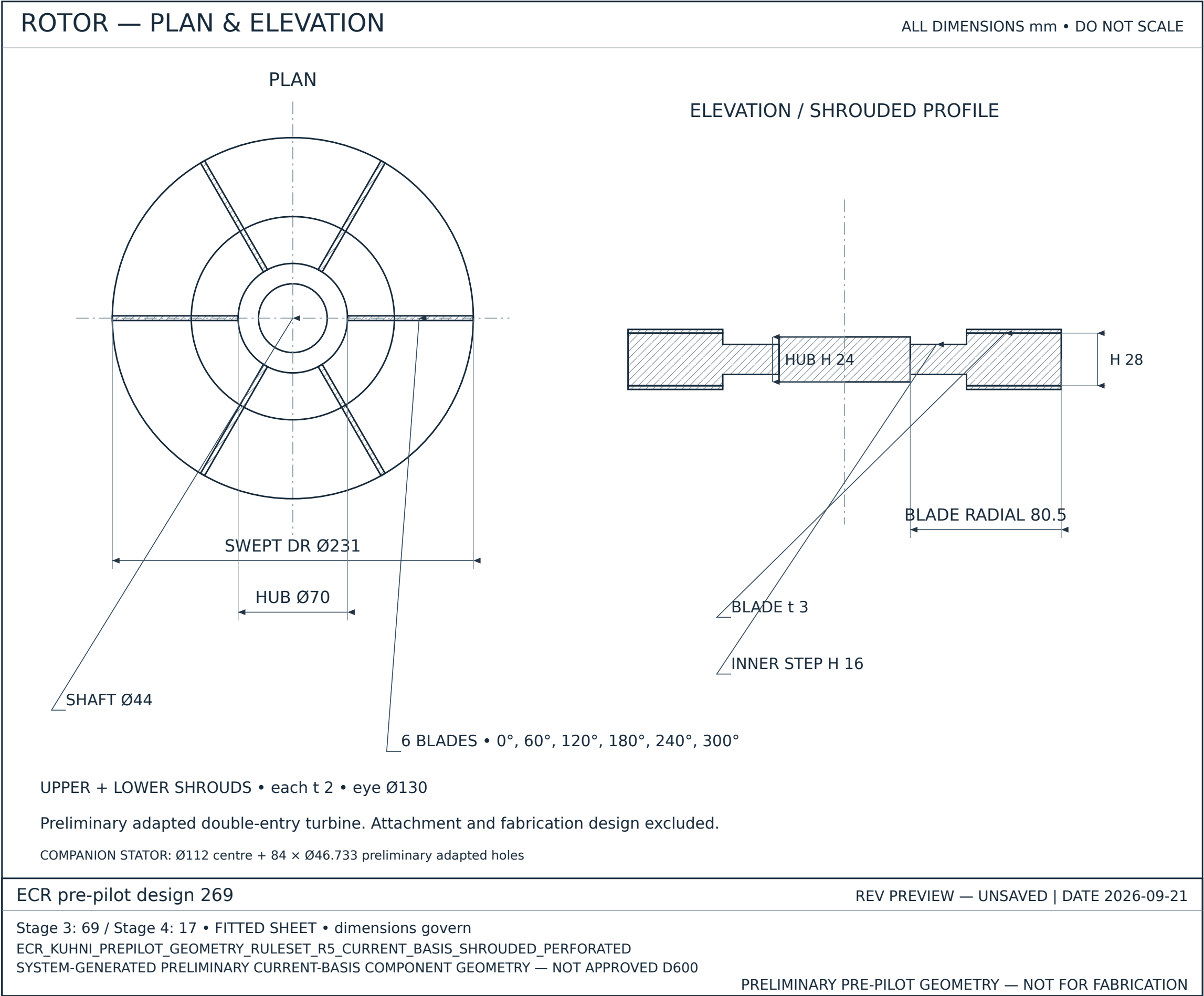
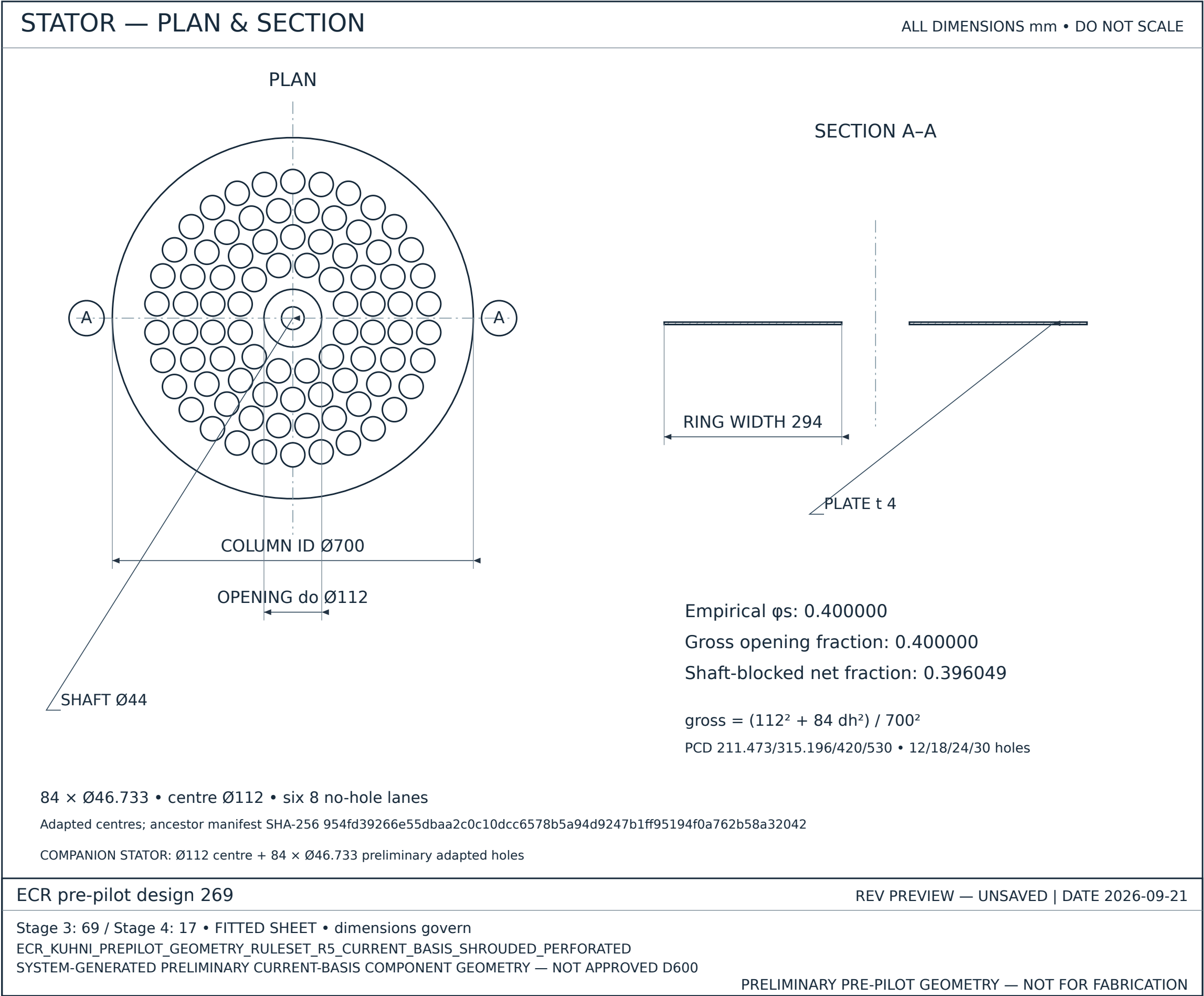










Geometry summary — frozen revision dimensions

PRELIMINARY PRE-PILOT GEOMETRY — NOT FOR FABRICATION

Geometrically complete pre-pilot layout only; not fabrication or mechanical qualification.

sourceHash: d69bc153d009196c806639257001dd8abf0914f572e4d1809cbf5f440e078138 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

stage3ResultId: 69 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

stage4ResultId: 17 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

columnDiameterM: 0.7 m | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

compartmentHeightM: 0.21 m | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

rotorDiameterRatio: 0.33 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

rotorDiameterM: 0.23099999999999998 m | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

statorFreeAreaRatio: 0.4 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

selectedRpm: 30 rpm | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

rpmMin: 30 rpm | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

rpmMax: 30 rpm | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

phaseConfiguration: rrbo-continuous-nmp-dispersed | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

compartmentCount: 20 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

requiredActiveHeightM: 4.2 m | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

installedActiveHeightM: 4.2 m | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

designNt: 7 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

sizingMethod: ADOPTED_COMPARTMENT_EFFICIENCY | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

designCompartmentEfficiency: 0.35 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

impliedInstalledHetsMPerTheoreticalStage: 0.6 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

rotorOffsetM: 0.105 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. hc/2; centred rotor.

bottomDisengagementM: 0.7 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. D; packaging allocation, not separation-time qualification.

topDisengagementM: 0.7 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. D; packaging allocation, not separation-time qualification.

bottomHeadDepthM: 0.175 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. D/4; 2:1 ellipsoidal internal envelope.

topHeadDepthM: 0.175 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. D/4; 2:1 ellipsoidal internal envelope.

headStraightFlangeM: 0 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Zero in R1 internal envelope model.

supportHeightM: 0.35 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. D/2; skirt below vessel datum.

skirtDiameterM: 0.7 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. D; centred cylindrical envelope.

skirtAccessWidthM: 0.21 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. 0.30D; access opening at azimuth zero.

skirtAccessHeightM: 0.175 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. 0.25D; access opening.

skirtAccessElevationM: -0.175 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. -0.25D; centre elevation.

lowerShaftSupportM: 0.5249999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. hB+LB/2.

upperShaftSupportM: 5.425 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. z1+LT/2.

supportArmWidthM: 0.013999999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. 0.02D; tangential width, three arms at 30/150/270 degrees.

supportArmDepthM: 0.027999999999999997 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. 0.04D; axial depth.

supportHousingDiameterM: 0.07699999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. 1.5ds; central envelope.

supportHousingHeightM: 0.027999999999999997 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. 0.04D; central envelope.

supportArmRadialStartM: 0.03849999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. 0.75ds; housing surface.

supportArmRadialEndM: 0.35 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. D/2; shell interface.

shaftLowerTerminationM: 0.5249999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Lower support centre.

shaftUpperTerminationM: 6.125 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. HV+D/4; drive engagement.

shaftLengthM: 5.6 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Upper minus lower termination.

sealElevationM: 5.95 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Top pole; coaxial with shaft.

sealDiameterM: 0.10266666666666666 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. 2ds; reserved housing, not product selection.

sealHeightM: 0.06999999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. 0.10D above top pole.

drivePedestalDiameterM: 0.35 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. D/2; coaxial reserved envelope.

drivePedestalHeightM: 0.175 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. D/4 above top pole.

driveBodyDiameterM: 0.35 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. D/2; reserved envelope.

driveBodyHeightM: 0.5249999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. 3D/4 above pedestal.

driveHeightM: 0.7 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. D including pedestal; not gearbox adequacy.

columnAreaM2: 0.38484510006474965 m² | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. $\pi D^2/4$.

shaftAreaM2: 0.0020696114270148756 m² | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. $\pi d_s^2/4$.

statorGrossOpenAreaM2: 0.15393804002589984 m² | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; recomputed from component solids; no upstream feedback.

statorNetOpenAreaM2: 0.15241750918156238 m² | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; recomputed from component solids; no upstream feedback.

grossFreeAreaRatio: 0.4 | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; recomputed from component solids; no upstream feedback.

shaftBlockedFreeAreaRatio: 0.39604897959183677 | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; recomputed from component solids; no upstream feedback.

calculatedFreeAreaRatio: 0.4 | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; recomputed from component solids; no upstream feedback.

rotorWallClearanceM: 0.2345 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; recomputed from component solids; no upstream feedback.

shaftOpeningClearanceM: 0.034 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; recomputed from component solids; no upstream feedback.

statorRadialWidthM: 0.294 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; recomputed from component solids; no upstream feedback.

rotorLowerClearanceM: 0.087 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; recomputed from component solids; no upstream feedback.

rotorUpperClearanceM: 0.087 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; recomputed from component solids; no upstream feedback.

activeStartM: 0.875 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. hB+LB.

activeEndM: 5.075 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. z0+HA.

topTangentM: 5.775 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. z1+LT.

vesselHeightM: 5.95 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. HA+2.5D.

overallHeightM: 7 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Support+vessel+drive = HA+4D.

flowArrangement: nmp-down-rrho-up | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Explicit application layout; continuity and upstream hydraulics unchanged.

headProfiles: 2:1 ellipsoidal | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Internal envelope only.

bladeAzimuthsDeg: 0, 60, 120, 180, 240, 300 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Six equally spaced blades; same clocking in every compartment.

supportAzimuthsDeg: 30, 150, 270 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Approved three-arm support clocking.

skirtAccessAzimuthDeg: 0 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Approved skirt access orientation.

designEnvelope: hc/D 0.20–0.30; DR/D 0.33–0.50; ϕ_s 0.20–0.40 | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class B. Perry 8e p15-83 → Pratt-Stevens; advisory, never clamp inherited geometry.

P01-elevationM: 0.595 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.

P01-boreM: 0.06999999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.

P01-azimuthDeg: 0 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.

P01-radialOffsetM: 0.35 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.

P01-outsideDiameterM: 0.08399999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.

P01-projectionM: 0.06999999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.

P02-elevationM: 0.45499999999999996 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P02-boreM: 0.06999999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P02-azimuthDeg: 180 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P02-radialOffsetM: 0.35 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P02-outsideDiameterM: 0.08399999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P02-projectionM: 0.06999999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P03-elevationM: 5.355 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P03-boreM: 0.06999999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P03-azimuthDeg: 180 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P03-radialOffsetM: 0.35 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

[illegible]

compartment17-rotorM: 4.34 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment18-bottomM: 4.445 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment18-topM: 4.654999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment18-rotorM: 4.55 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment19-bottomM: 4.654999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment19-topM: 4.865 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment19-rotorM: 4.76 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment20-bottomM: 4.865 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment20-topM: 5.075 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment20-rotorM: 4.97 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

shaftDiameterM: 0.044 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Retained component-basis shaft diameter.

statorThicknessM: 0.004 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Retained component-basis plate thickness.

bladeCount: 6 | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Retained component-basis six-blade architecture.

bladeHeightM: 0.028 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Retained component-basis outer paddle height.

bladeInnerSegmentHeightM: 0.016 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Retained component-basis inner web height.

bladeOuterSegmentHeightM: 0.028 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Retained component-basis outer paddle height.

bladeThicknessM: 0.003 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Retained component-basis blade thickness.

hubDiameterM: 0.07 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Retained component-basis hub diameter.

hubHeightM: 0.024 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Retained component-basis hub height.

rotorAxialEnvelopeM: 0.032 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Retained component-basis total shrouded rotor height.

bladeRadialLengthM: 0.08049999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Inherited rotor radius minus retained 35mm hub radius; preliminary adaptation.

bladeInnerSegmentLengthM: 0.03 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Retained component-basis web radial interval, 35 to 65 mm radius.

bladeOuterSegmentLengthM: 0.05049999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Inherited rotor radius minus retained 65mm eye radius; preliminary adaptation.

shroudThicknessM: 0.002 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Each Retained component-basis upper/lower shroud.

shroudInnerDiameterM: 0.13 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Retained component-basis eye diameter.

shroudOuterDiameterM: 0.23099999999999998 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Inherited rotor OD; preliminary adaptation, not Retained component-basis D600.

statorOpeningDiameterM: 0.112 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Retained component-basis central opening (130 - 2×9).

statorHoleCount: 84 | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Retained component-basis exact perforation count.

statorHoleDiameterM: 0.046733285782191686 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. sqrt((inherited phi × inherited D² - 0.112²)/84); retained row angles and count; minimum outward row movement only to resolve 8mm-lane interference.

statorLaneWidthM: 0.008 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Six retained radial no-hole lanes.

rotorConstruction: approved-double-entry-shrouded-turbine | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Retained component-basis component architecture identifier.

statorConstruction: approved-perforated-stator | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED; Class C. Retained component-basis perforated plate architecture identifier.

Compartment and internals schedules

- Compartment 1: bottom 0.875 m; top 1.085 m; rotor 0.98 m; stator 1.085 m
- Compartment 2: bottom 1.085 m; top 1.295 m; rotor 1.19 m; stator 1.295 m
- Compartment 3: bottom 1.295 m; top 1.505 m; rotor 1.4 m; stator 1.505 m
- Compartment 4: bottom 1.505 m; top 1.714999999999999 m; rotor 1.609999999999999 m; stator 1.714999999999999 m
- Compartment 5: bottom 1.714999999999999 m; top 1.925 m; rotor 1.819999999999998 m; stator 1.925 m
- Compartment 6: bottom 1.925 m; top 2.135 m; rotor 2.03000000000000002 m; stator 2.135 m
- Compartment 7: bottom 2.135 m; top 2.344999999999998 m; rotor 2.24 m; stator 2.344999999999998 m
- Compartment 8: bottom 2.344999999999998 m; top 2.554999999999997 m; rotor 2.45 m; stator 2.554999999999997 m
- Compartment 9: bottom 2.554999999999997 m; top 2.764999999999997 m; rotor 2.66 m; stator 2.764999999999997 m
- Compartment 10: bottom 2.764999999999997 m; top 2.975 m; rotor 2.87 m; stator 2.975 m
- Compartment 11: bottom 2.975 m; top 3.185 m; rotor 3.08 m; stator 3.185 m
- Compartment 12: bottom 3.185 m; top 3.395 m; rotor 3.29 m; stator 3.395 m
- Compartment 13: bottom 3.395 m; top 3.605 m; rotor 3.5 m; stator 3.605 m
- Compartment 14: bottom 3.605 m; top 3.815 m; rotor 3.71 m; stator 3.815 m
- Compartment 15: bottom 3.815 m; top 4.025 m; rotor 3.92 m; stator 4.025 m
- Compartment 16: bottom 4.025 m; top 4.234999999999999 m; rotor 4.13 m; stator 4.234999999999999 m
- Compartment 17: bottom 4.234999999999999 m; top 4.445 m; rotor 4.34 m; stator 4.445 m
- Compartment 18: bottom 4.445 m; top 4.654999999999999 m; rotor 4.55 m; stator 4.654999999999999 m
- Compartment 19: bottom 4.654999999999999 m; top 4.865 m; rotor 4.76 m; stator 4.865 m
- Compartment 20: bottom 4.865 m; top 5.075 m; rotor 4.97 m; stator 5.075 m
- R | Preliminary double-entry shrouded six-blade turbine; retained 130mm eye | count 20 | diameter 0.2309999999999998 m | thickness 0.032 m | elevations 0.98, 1.19, 1.4, 1.609999999999999, 1.819999999999998, 2.03000000000000002, 2.24, 2.45, 2.66, 2.87, 3.08, 3.29, 3.5, 3.71, 3.92, 4.13, 4.34, 4.55, 4.76, 4.97 m
- S | Preliminary perforated stator; retained 112mm centre and 84-hole architecture | count 21 | diameter 0.7 m | thickness 0.004 m | elevations 0.875, 1.085, 1.295, 1.505, 1.714999999999999, 1.925, 2.135, 2.344999999999998, 2.554999999999997, 2.764999999999997, 2.975, 3.185, 3.395, 3.605, 3.815, 4.025, 4.234999999999999, 4.445, 4.654999999999999, 4.865, 5.075 m
- SH | Continuous shaft envelope; mechanical qualification excluded | count 1 | diameter 0.0513333333333333 m | thickness 0.0513333333333333 m | elevations 0.524999999999999, 6.125 m
- SUP | Three-arm shaft supports, azimuths 30/150/270 deg | count 2 | diameter 0.7 m | thickness 0.0279999999999997 m | elevations 0.524999999999999, 5.425 m

Preliminary nozzle / connection schedule

- P01 | RRBO feed | bottom | elevation 0.595 m | bore 0.0699999999999999 m | azimuth 0 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.0839999999999999 m; projection 0.0699999999999999 m; radial offset 0.35 m
- P02 | NMP-rich extract outlet | bottom | elevation 0.4549999999999996 m | bore 0.0699999999999999 m | azimuth 180 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.0839999999999999 m; projection 0.0699999999999999 m; radial offset 0.35 m
- P03 | NMP feed | top | elevation 5.355 m | bore 0.0699999999999999 m | azimuth 180 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.0839999999999999 m; projection 0.0699999999999999 m; radial offset 0.35 m
- P04 | RRBO-rich raffinate outlet | top | elevation 5.495 m | bore 0.0699999999999999 m | azimuth 0 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.0839999999999999 m; projection 0.0699999999999999 m; radial offset 0.35 m
- A01 | Low-shell flush | bottom | elevation 0.3149999999999995 m | bore 0.02799999999999997 m | azimuth 270 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/25; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.0336 m; projection 0.0699999999999999 m; radial offset 0.35 m
- A02 | High-shell equalization | top | elevation 5.635 m | bore 0.02799999999999997 m | azimuth 270 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/25; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.0336 m; projection 0.0699999999999999 m; radial offset 0.35 m
- S01 | Lower sample connection | bottom | elevation 0.734999999999999 m | bore 0.01749999999999998 m | azimuth 90 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/40; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.02099999999999998 m; projection 0.0699999999999999 m; radial offset 0.35 m
- S02 | Upper sample connection | top | elevation 5.215 m | bore 0.01749999999999998 m | azimuth 90 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/40; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.02099999999999998 m; projection 0.0699999999999999 m; radial offset 0.35 m
- I01 | Lower instrument connection | bottom | elevation 0.385 m | bore 0.01749999999999998 m | azimuth 45 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.02099999999999998 m; projection 0.0699999999999999 m; radial offset 0.35 m
- I02 | Upper instrument connection | top | elevation 5.285 m | bore 0.01749999999999998 m | azimuth 45 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.02099999999999998 m; projection 0.0699999999999999 m; radial offset 0.35 m
- D01 | Bottom-head drain | bottom-head | elevation 0 m | bore 0.02799999999999997 m | azimuth N/A axial deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: bottom-pole axial drain; azimuth N/A. | axis down; OD 0.0336 m; projection 0.0699999999999999 m; radial offset 0 m
- V01 | Top-head crown-region vent | top-head | elevation 5.915 m | bore 0.02799999999999997 m | azimuth 315 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; C: r=0.30D, ellipsoidal surface z=zT+0.20D; not absolute-high-point or vent-duty qualification. | axis up; OD 0.0336 m; projection 0.0699999999999999 m; radial offset 0.21 m

Validation, assumptions and unresolved items

- PASS: Frozen sources must be identified, current and compatible.
- PASS: columnDiameterM must be finite and positive.
- PASS: rotorDiameterM must be finite and positive.
- PASS: rotorDiameterRatio must be finite and positive.
- PASS: compartmentHeightM must be finite and positive.
- PASS: compartmentCount must be finite and positive.
- PASS: installedActiveHeightM must be finite and positive.
- PASS: requiredActiveHeightM must be finite and positive.
- PASS: designNt must be finite and positive.
- PASS: selectedRpm must be finite and positive.
- PASS: rpmMin must be finite and positive.
- PASS: rpmMax must be finite and positive.
- PASS: statorFreeAreaRatio must be finite and positive.
- PASS: Positive integer count; drawing capacity 1000 compartments.
- PASS: Frozen count × pitch must equal frozen installed height; no repair.
- PASS: Frozen adopted sizing basis must be Nt=7 and compartment efficiency=0.35.
- PASS: Frozen count must equal ceil(Nt / adopted compartment efficiency).
- PASS: Frozen required and installed heights must both equal count × pitch.
- PASS: Implied installed HETS must equal installed height / Nt and remains diagnostic only.
- PASS: Frozen rotor diameter/ratio must agree and fit column.
- PASS: Frozen selected RPM must lie inside frozen window.
- PASS: Frozen phase continuity is required.
- PASS: Empirical free-area input must be between zero and one.
- PASS: columnDiameterM must be finite.
- PASS: compartmentHeightM must be finite.
- PASS: rotorDiameterRatio must be finite.
- PASS: rotorDiameterM must be finite.
- PASS: statorFreeAreaRatio must be finite.
- PASS: selectedRpm must be finite.
- PASS: rpmMin must be finite.
- PASS: rpmMax must be finite.
- PASS: compartmentCount must be finite.
- PASS: requiredActiveHeightM must be finite.
- PASS: installedActiveHeightM must be finite.
- PASS: designNt must be finite.
- PASS: designCompartmentEfficiency must be finite.
- PASS: impliedInstalledHetsMPerTheoreticalStage must be finite.
- PASS: shaftDiameterM must be finite.
- PASS: statorThicknessM must be finite.
- PASS: bladeCount must be finite.
- PASS: bladeHeightM must be finite.
- PASS: hubDiameterM must be finite.
- PASS: hubHeightM must be finite.
- PASS: bladeThicknessM must be finite.
- PASS: bladeRadialLengthM must be finite.
- PASS: bladeInnerSegmentLengthM must be finite.
- PASS: bladeOuterSegmentLengthM must be finite.
- PASS: bladeInnerSegmentHeightM must be finite.
- PASS: bladeOuterSegmentHeightM must be finite.
- PASS: rotorOffsetM must be finite.

PASS: bottomDisengagementM must be finite.
PASS: topDisengagementM must be finite.
PASS: bottomHeadDepthM must be finite.
PASS: topHeadDepthM must be finite.
PASS: headStraightFlangeM must be finite.
PASS: supportHeightM must be finite.
PASS: skirtDiameterM must be finite.
PASS: skirtAccessWidthM must be finite.
PASS: skirtAccessHeightM must be finite.
PASS: skirtAccessElevationM must be finite.
PASS: lowerShaftSupportM must be finite.
PASS: upperShaftSupportM must be finite.
PASS: supportArmWidthM must be finite.
PASS: supportArmDepthM must be finite.
PASS: supportHousingDiameterM must be finite.
PASS: supportHousingHeightM must be finite.
PASS: supportArmRadialStartM must be finite.
PASS: supportArmRadialEndM must be finite.
PASS: shaftLowerTerminationM must be finite.
PASS: shaftUpperTerminationM must be finite.
PASS: shaftLengthM must be finite.
PASS: sealElevationM must be finite.
PASS: sealDiameterM must be finite.
PASS: sealHeightM must be finite.
PASS: drivePedestalDiameterM must be finite.
PASS: drivePedestalHeightM must be finite.
PASS: driveBodyDiameterM must be finite.
PASS: driveBodyHeightM must be finite.
PASS: driveHeightM must be finite.
PASS: statorOpeningDiameterM must be finite.
PASS: columnAreaM2 must be finite.
PASS: shaftAreaM2 must be finite.
PASS: statorGrossOpenAreaM2 must be finite.
PASS: statorNetOpenAreaM2 must be finite.
PASS: grossFreeAreaRatio must be finite.
PASS: shaftBlockedFreeAreaRatio must be finite.
PASS: calculatedFreeAreaRatio must be finite.
PASS: rotorWallClearanceM must be finite.
PASS: shaftOpeningClearanceM must be finite.
PASS: statorRadialWidthM must be finite.
PASS: rotorAxialEnvelopeM must be finite.
PASS: rotorLowerClearanceM must be finite.
PASS: rotorUpperClearanceM must be finite.
PASS: activeStartM must be finite.
PASS: activeEndM must be finite.
PASS: topTangentM must be finite.
PASS: vesselHeightM must be finite.
PASS: overallHeightM must be finite.
PASS: skirtAccessAzimuthDeg must be finite.
PASS: P01-elevationM must be finite.
PASS: P01-boreM must be finite.
PASS: P01-azimuthDeg must be finite.

PASS: P01-radialOffsetM must be finite.
PASS: P01-outsideDiameterM must be finite.
PASS: P01-projectionM must be finite.
PASS: P02-elevationM must be finite.
PASS: P02-boreM must be finite.
PASS: P02-azimuthDeg must be finite.
PASS: P02-radialOffsetM must be finite.
PASS: P02-outsideDiameterM must be finite.
PASS: P02-projectionM must be finite.
PASS: P03-elevationM must be finite.
PASS: P03-boreM must be finite.
PASS: P03-azimuthDeg must be finite.
PASS: P03-radialOffsetM must be finite.
PASS: P03-outsideDiameterM must be finite.
PASS: P03-projectionM must be finite.
PASS: P04-elevationM must be finite.
PASS: P04-boreM must be finite.
PASS: P04-azimuthDeg must be finite.
PASS: P04-radialOffsetM must be finite.
PASS: P04-outsideDiameterM must be finite.
PASS: P04-projectionM must be finite.
PASS: A01-elevationM must be finite.
PASS: A01-boreM must be finite.
PASS: A01-azimuthDeg must be finite.
PASS: A01-radialOffsetM must be finite.
PASS: A01-outsideDiameterM must be finite.
PASS: A01-projectionM must be finite.
PASS: A02-elevationM must be finite.
PASS: A02-boreM must be finite.
PASS: A02-azimuthDeg must be finite.
PASS: A02-radialOffsetM must be finite.
PASS: A02-outsideDiameterM must be finite.
PASS: A02-projectionM must be finite.
PASS: S01-elevationM must be finite.
PASS: S01-boreM must be finite.
PASS: S01-azimuthDeg must be finite.
PASS: S01-radialOffsetM must be finite.
PASS: S01-outsideDiameterM must be finite.
PASS: S01-projectionM must be finite.
PASS: S02-elevationM must be finite.
PASS: S02-boreM must be finite.
PASS: S02-azimuthDeg must be finite.
PASS: S02-radialOffsetM must be finite.
PASS: S02-outsideDiameterM must be finite.
PASS: S02-projectionM must be finite.
PASS: I01-elevationM must be finite.
PASS: I01-boreM must be finite.
PASS: I01-azimuthDeg must be finite.
PASS: I01-radialOffsetM must be finite.
PASS: I01-outsideDiameterM must be finite.
PASS: I01-projectionM must be finite.
PASS: I02-elevationM must be finite.

PASS: I02-boreM must be finite.
PASS: I02-azimuthDeg must be finite.
PASS: I02-radialOffsetM must be finite.
PASS: I02-outsideDiameterM must be finite.
PASS: I02-projectionM must be finite.
PASS: D01-elevationM must be finite.
PASS: D01-boreM must be finite.
PASS: D01-radialOffsetM must be finite.
PASS: D01-outsideDiameterM must be finite.
PASS: D01-projectionM must be finite.
PASS: D01-surfaceEdgeElevationMinM must be finite.
PASS: D01-surfaceEdgeElevationMaxM must be finite.
PASS: Entire neck cylinder fits ellipsoidal head.
PASS: V01-elevationM must be finite.
PASS: V01-boreM must be finite.
PASS: V01-azimuthDeg must be finite.
PASS: V01-radialOffsetM must be finite.
PASS: V01-outsideDiameterM must be finite.
PASS: V01-projectionM must be finite.
PASS: V01-surfaceEdgeElevationMinM must be finite.
PASS: V01-surfaceEdgeElevationMaxM must be finite.
PASS: compartment1-bottomM must be finite.
PASS: compartment1-topM must be finite.
PASS: compartment1-rotorM must be finite.
PASS: compartment2-bottomM must be finite.
PASS: compartment2-topM must be finite.
PASS: compartment2-rotorM must be finite.
PASS: compartment3-bottomM must be finite.
PASS: compartment3-topM must be finite.
PASS: compartment3-rotorM must be finite.
PASS: compartment4-bottomM must be finite.
PASS: compartment4-topM must be finite.
PASS: compartment4-rotorM must be finite.
PASS: compartment5-bottomM must be finite.
PASS: compartment5-topM must be finite.
PASS: compartment5-rotorM must be finite.
PASS: compartment6-bottomM must be finite.
PASS: compartment6-topM must be finite.
PASS: compartment6-rotorM must be finite.
PASS: compartment7-bottomM must be finite.
PASS: compartment7-topM must be finite.
PASS: compartment7-rotorM must be finite.
PASS: compartment8-bottomM must be finite.
PASS: compartment8-topM must be finite.
PASS: compartment8-rotorM must be finite.
PASS: compartment9-bottomM must be finite.
PASS: compartment9-topM must be finite.
PASS: compartment9-rotorM must be finite.
PASS: compartment10-bottomM must be finite.
PASS: compartment10-topM must be finite.
PASS: compartment10-rotorM must be finite.
PASS: compartment11-bottomM must be finite.

[illegible]

ASSUMPTION: Approved component source: approved-component-manifest.json canonical SHA-256 954fd39266e55dbaa2c0c10dcc6578b5a94d9247b1ff95194f0a762b58a32042.

ASSUMPTION: PRELIMINARY INTERFACE HOLD: retain the existing vessel shell, shaft-support elevations/wall interfaces, drive/seal envelopes and nozzle arrangement pending mechanical interface review.

ASSUMPTION: Component geometry approval does not qualify support-wall attachments, stator mounting, shaft/hub attachment, strength, vibration or fabrication.

ASSUMPTION: R5 PRELIMINARY ADAPTATION, NOT APPROVED D600: manifest identifies retained component ancestry, not approval of this generated assembly.

ASSUMPTION: Rotor/shroud outer radius and stator hole diameter follow inherited rotor OD and gross free area. Row radii change only by the minimum outward displacement necessary to clear the retained 8mm no-hole lanes; count, angles, 130mm eye, 112mm centre and 9mm overlap remain fixed. No 95% turning-area rule.

ASSUMPTION: Preliminary stator row radii (m): 0.1 → 0.10573658858935844; 0.155 → 0.1575982153039753; 0.21 → 0.21; 0.265 → 0.265. Changed radii resolve actual lane interference only; no hydraulic equivalence asserted.

ASSUMPTION: Retained component thicknesses are preliminary geometric assumptions, not strength-qualified thicknesses. Materials, shell thickness, support/drive adequacy and fabrication qualification remain unverified.

ASSUMPTION: Inherited RPM 30 is a zero-width selected point, not a demonstrated operating window.

OUTSIDE ENGINEERING SCOPE: Pressure-vessel shell/head thickness and code design

OUTSIDE ENGINEERING SCOPE: Shaft strength, deflection, critical speed

OUTSIDE ENGINEERING SCOPE: Bearing/seal ratings, drive power/torque and support structural qualification

OUTSIDE ENGINEERING SCOPE: Welds, keys, fasteners, flanges, reinforcement and external pipe routing

OUTSIDE ENGINEERING SCOPE: Separation residence-time, connection hydraulic capacity and operational venting qualification

Engineering notes: ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_ADOPTED_COMPARTMENT_EFFICIENCY_ETA0.35; system-generated; no normal-user construction inputs.

Revision notes: Read-only current-authority export. Revision 0 denotes an unsaved preview; no business revision was created.

Stage 3 result: 69; Stage 4 result: 17

Source SHA-256: d69bc153d009196c806639257001dd8abf0914f572e4d1809cbf5f440e078138

PRELIMINARY — NOT FOR FABRICATION. Mechanical verification, pressure-vessel design, shaft strength/deflection/critical speed, bearing/seal selection, drive adequacy and structural supports are outside this package.